

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Drill	1	hr

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9702/33

May/June 2012

2 hours

Additional Materials: As listed in the Confidential Instructions.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

You are reminded of the need for good English and clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

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1	
2	
Total	

This document consists of **10** printed pages and **2** blank pages.

You may not need to use all of the materials provided.

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- 1** In this experiment, you will investigate how the extension of an arrangement of springs depends on the loads applied to it.

(a) (i) Set up the apparatus as shown in Fig. 1.1.

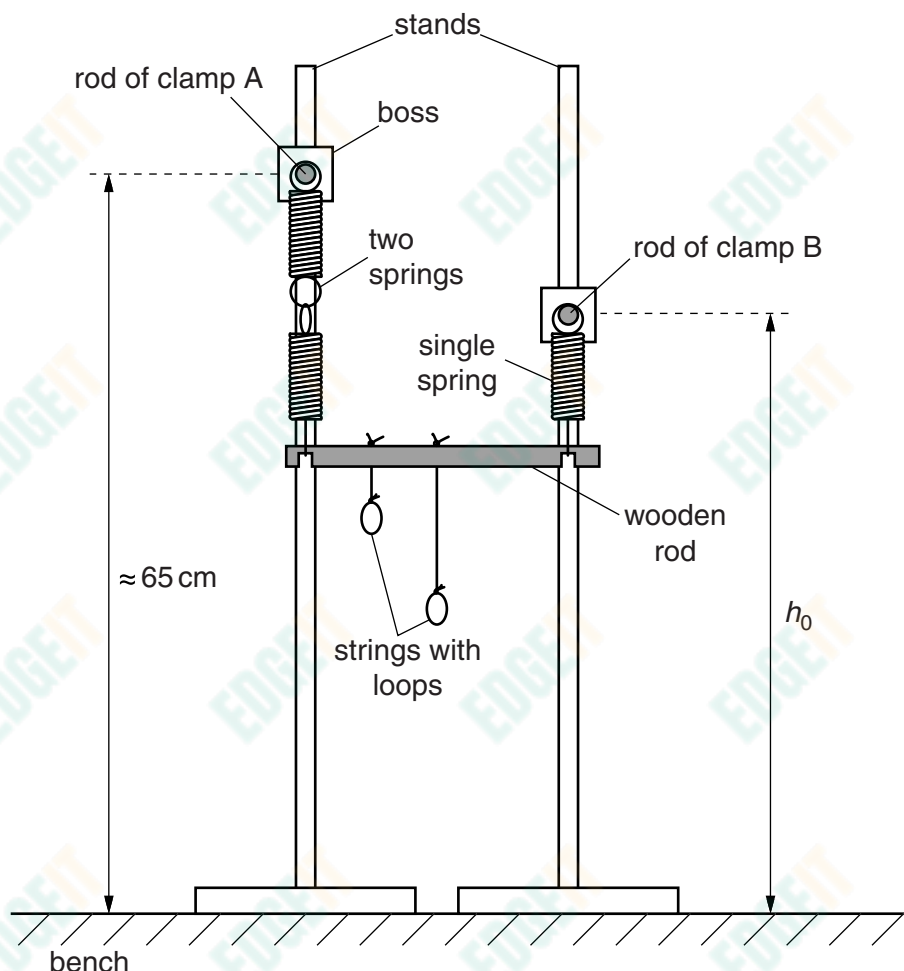


Fig. 1.1

Place the rod of clamp A, supporting the two springs, approximately 65 cm above the bench.

Once fixed do not change the position of the rod of clamp A throughout the experiment.

Suspend the wooden rod, with strings attached, using the springs as shown. Adjust the position of the rod of clamp B until the wooden rod is horizontal. Ensure the springs hang vertically.

- (ii)** Measure and record the distance h_0 between the centre of the rod of clamp B (supporting the single spring) and the bench.

$h_0 = \dots\dots\dots$ m [1]

- (b) (i) Add masses to the wooden rod as shown in Fig. 1.2. The 100 g mass hanger should be attached to the longer, central string.

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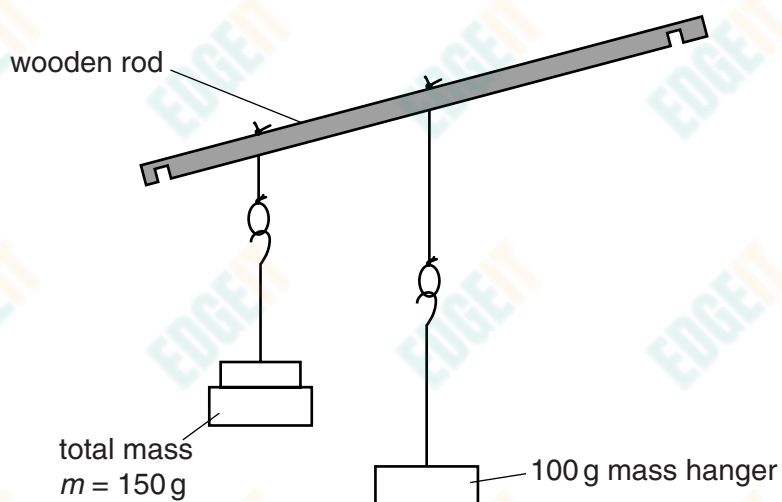


Fig. 1.2

- (ii) Adjust the height of the rod of clamp B until the wooden rod is horizontal.
- (iii) Measure and record the distance h between the centre of the rod of clamp B and the bench, as shown in Fig. 1.3.

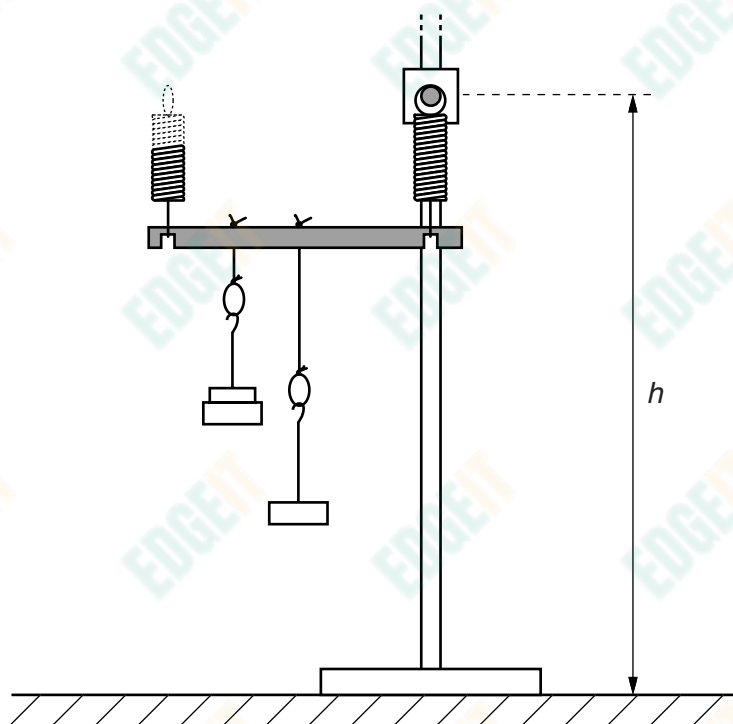


Fig. 1.3

$h = \dots\dots\dots$ [1]

(iv) Calculate the value of $(h_0 - h)$.

$$(h_0 - h) = \dots\dots\dots$$

- (c) By increasing the mass m , repeat **(b)(ii)**, **(b)(iii)** and **(b)(iv)** until you have five sets of values of m and h . Do not change the mass attached to the longer, central string.

Include values of $\frac{(h_0 - h)}{m}$ and $\frac{1}{m}$ in your table.

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[10]

- (d) (i) Plot a graph of $\frac{(h_0 - h)}{m}$ on the y-axis against $\frac{1}{m}$ on the x-axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

- (iii) Determine the gradient and y-intercept of this line.

$$\text{gradient} = \dots\dots\dots$$

$$\text{y-intercept} = \dots\dots\dots$$

[2]

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- (e) The quantities h and m are related by the equation

$$\frac{(h_0 - h)}{m} = \frac{P}{m} + Q$$

where P and Q are constants.

Use your answers in (d)(iii) to determine the values of P and Q .
Give appropriate units.

$P =$

$Q =$

[2]

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- 2** In this experiment, you will investigate how the cooling rate of a hot liquid depends on the surface area of the liquid exposed to air.

- (a)**
- (i)** Pour cold water into the beaker up to the 200 ml mark.
 - (ii)** Pour the water into the cup and use the pen to place a mark on the inside surface of the cup, level with the water surface.
 - (iii)** Empty out the cold water.
 - (iv)** Repeat **(a)(i)**, **(a)(ii)** and **(a)(iii)** for the bowl.

- (b)**
- (i)** Pour boiling water into the cup up to the mark.
 - (ii)** When the temperature of the water falls to approximately 75 °C, start the stopwatch.

Record this starting temperature θ_0 .

$\theta_0 = \dots\dots\dots$ [1]

- (iii)** After two minutes, measure and record the temperature θ .

$\theta = \dots\dots\dots$ [1]

- (iv)** Calculate the change in temperature $\Delta\theta = (\theta_0 - \theta)$.

$\Delta\theta = \dots\dots\dots$ [1]

- (c)**
- (i)** Measure and record the diameter d of the water surface.

$d = \dots\dots\dots$ [1]

- (ii) Estimate the percentage uncertainty in your value of d .

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percentage uncertainty =[1]

- (d) Repeat (b) and (c)(i) for the bowl.

$\theta_0 =$

$\theta =$

$\Delta\theta = (\theta_0 - \theta) =$

$d =$

[4]

- (e) It is suggested that the relationship between $\Delta\theta$ and d is

$$\Delta\theta = k d^2$$

where k is a constant.

- (i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$

[1]

- (ii) Justify the number of significant figures that you have given for your values of k .

.....

.....

..... [1]

- (iii) Explain whether your results in (e)(i) support the suggested relationship.

.....

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..... [1]

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- (f) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

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1.
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2.
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3.
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4.
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[4]



- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

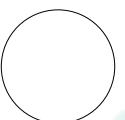
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[4]



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